

$$\mathcal{K}_1 : I^1 = (1, 2)$$

K_{11}^1	K_{12}^1
K_{21}^1	K_{22}^1

$$\mathcal{K}_2 : I^2 = (2, 3)$$

K_{11}^2	K_{12}^2
K_{21}^2	K_{22}^2

$$\mathcal{K}_3 : I^3 = (3, 4)$$

K_{11}^3	K_{12}^3
K_{21}^3	K_{22}^3



K_{11}^1	K_{12}^1	0	0
K_{21}^1	$K_{22}^1 + K_{11}^2$	K_{12}^2	0
0	K_{21}^2	$K_{22}^2 + K_{11}^3$	K_{12}^3
0	0	K_{21}^3	K_{22}^3

shared entries accumulate contributions from adjacent elements